Project 2

Open-Source AI Practice
(Deep Learning Programming Practice)

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Schedule

SUNG KYUN KWAN UNIVERSITY(SKKU)

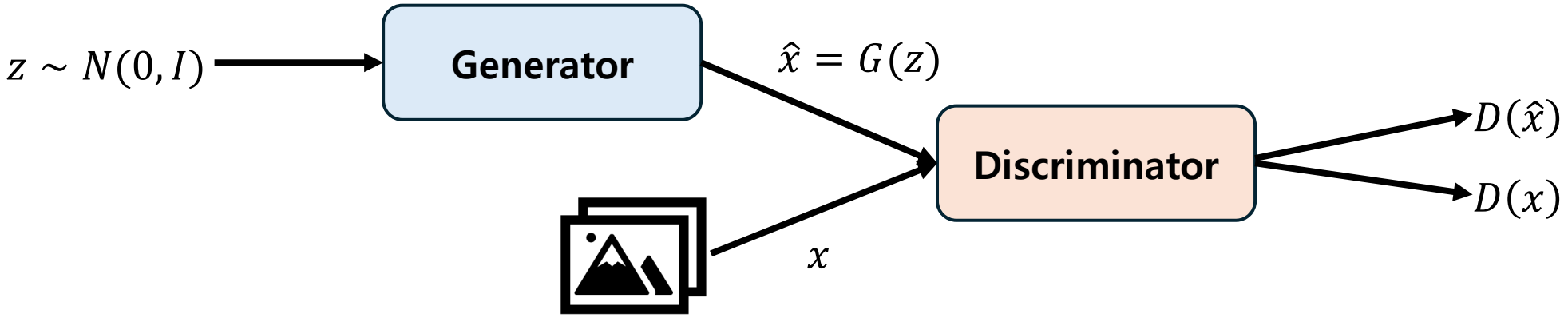
Date	Notes	Date	Notes
3. 4.	Orientation	4. 29.	**Project 1 QA**
3. 11.	**Tensor** **Assignment 02**	5. 6.	**Multi-task Learning** **Assignment 07**
3. 18.	Gas accident (no class)	5. 13.	Project 2 (~6/9) Announcement
3. 25.	**Module** **Assignment 03**	5. 20.	**Model Efficiency** **Assignment 08**
4. 1.	**Layer** **Assignment 04**	5. 27.	**Project 2 QA**
4. 8.	**Training** **Assignment 05**	6. 3.	Election day (no class)
4. 15.	Project 1 (~5/5) Announcement	6. 10.	**LLM Prompting** **Assignment 09**
4. 22.	**Optimization/Regularization** **Assignment 06**	6. 17.	Best project presentation

Project 2: Image Generation

- **Image Generation using GAN**
  - Generative adversarial network (GAN) is a model that generates images from noise.
  - GAN consists of Generator (G) and Discriminator (D).

  - GAN uses adversarial loss:

1. D should increase the D score for real images and decrease the D score for fake images.
2. G should increase the D score for fake (generated) images.



Project 2: Task

- **Generate high-quality and realistic human faces**
  - Target: 1024x1024 image generation (FFHQ dataset)

Project 2: FFHQ-1024

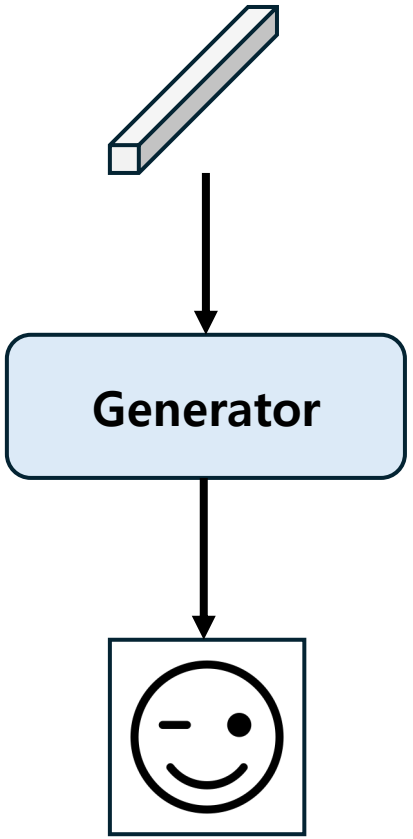
- **FFHQ-1024**
  - The dataset is high-quality, curated human face dataset [link].
  - **Data split will be announced.**
    - **50k – training**
    - **10k – valid**
    - **10k – test**
  - **DO NOT USE VALID or TEST for training**
  - Because the original images are too large (approx. 89GB for all), we will provide the **JPEG-compressed version** of the images.
    - This will cause JPG artifacts, but it is okay for this project.
    - Training and evaluation will both use the same JPG images.
  - Do not use external datasets.

Project 2: Model

- **Generator**
  - The total parameter size should be **under 40M**.
  - **Input: 512-dim random vector.**
  - **Output: 3x1024x1024 RGB image.**

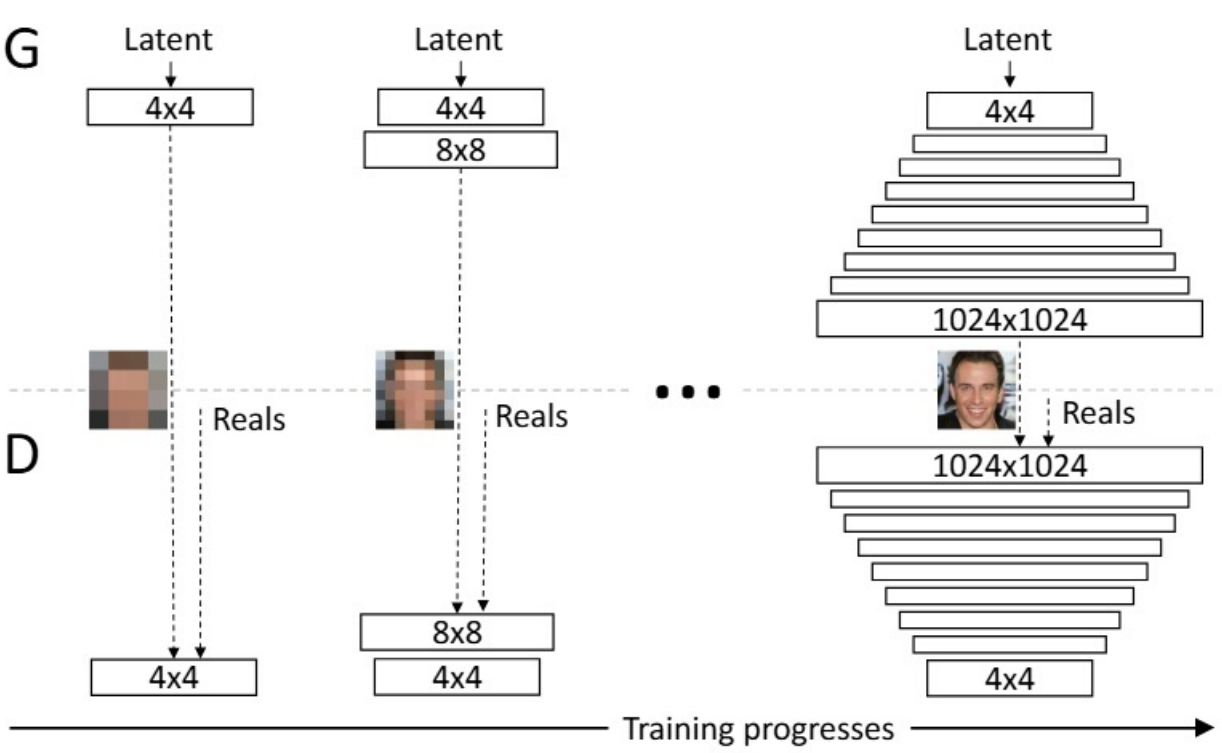
- **Discriminator**
  - No strict requirements.
  - D should be trained together with G.

- Useful references
  - BigGAN [paper]
  - SA-GAN [paper]
  - StyleGAN v2 [paper]
  - StyleGAN v3 [demo / paper]



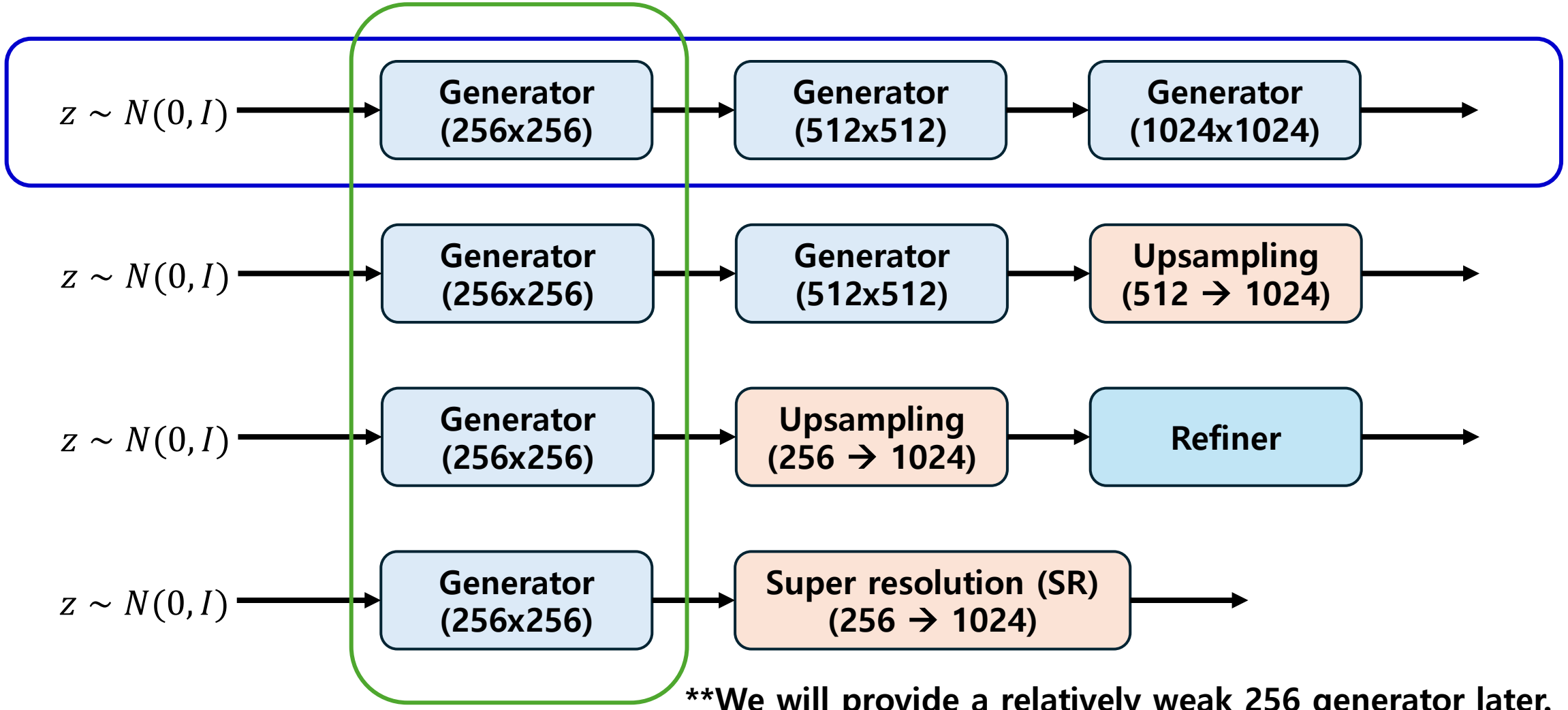
Project 2: Progressive Training

- **Progressive Growing**
  - Training a high-resolution image from scratch is challenging.
  - Progressive Growing (PG-GAN) [paper] starts from low-resolution images and progressively increases the resolution.
  - In this project, we may adopt this strategy to 256 → 512 → 1024.



Project 2: High-Resolution Strategy

- **High-resolution strategies**



**We will provide a relatively weak 256 generator later.

Project 2: Library

- **Library**
  - You can use:
    - AI: **PyTorch** and **TorchVision**
    - Vision: OpenCV, Pillow, Scikit-Image, Matplotlib, etc.
    - Tracking: WandB
    - Evaluation: **pytorch-fid**
  - Do not use:
    - HuggingFace libraries (Datasets, TIMM, Hub, Evaluate, etc.)
    - 3rd party libraries (PyTorch lightning, Accelerate, Albumentations, etc.) that support modeling, data processing, augmentation, training, or evaluation.
    - **DO NOT START FROM EXISTING REPOSITORIES / DO NOT USE PRETRAINED MODELS**
    - If you are not sure, ask professor ASAP.
- **AI Usage**
  - You are encouraged to get help from ChatGPT, Claude Code, or Gemini.
    - In your report, briefly describe which AI tools you used and how they helped you.

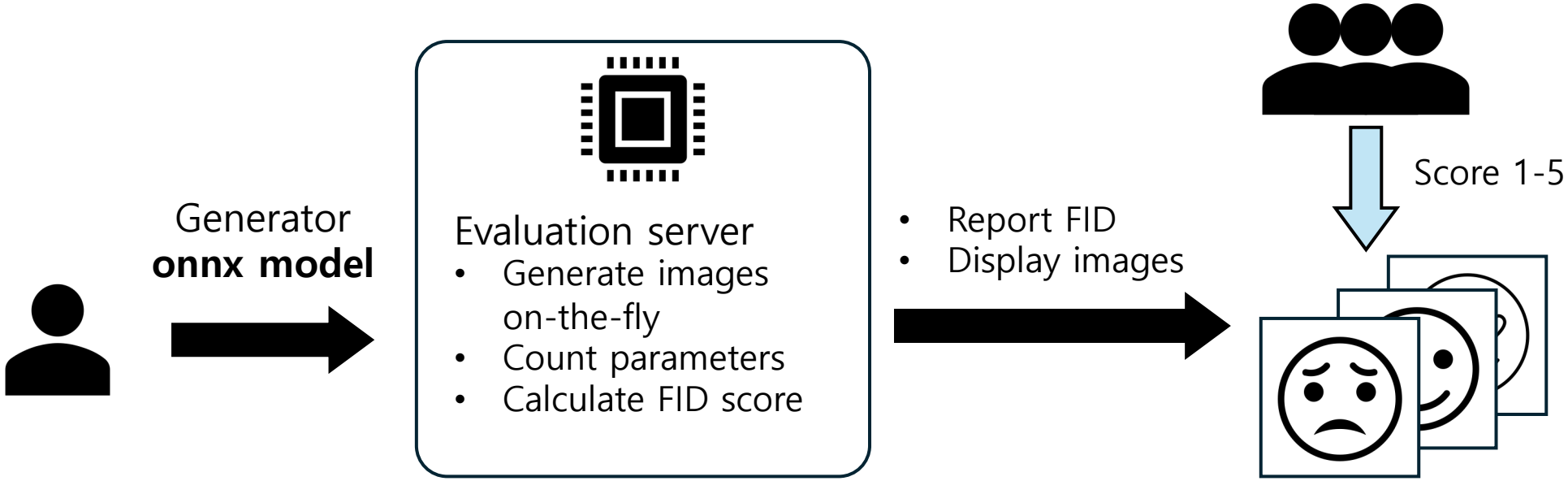
Project 2: Score

- **The total score will consider both quantitative and qualitative measures.**
  - Quantitative
    - **FID score** (performance) (**10 points**)
    - Parameter count (**hard threshold: G under 40M**) (**5 points**)
  - Qualitative
    - Professor's quality assessment (**5 points**)
    - Students' quality assessment (**5 points**)
  - Code quality (**5 points**)
  - Report quality (**5 points**)

Project 2: Score

- **Code quality (not limited to):**
  - Clear module structure
  - Reproducible training and evaluation scripts
  - Proper checkpoint loading and saving
  - Readable implementation
  - No hard-coded test-set assumptions
  - Try to use many things learned in class.
- **Report quality (not limited to):**
  - Model architecture
  - Training recipe
  - Validation results
  - Ablation or trial history
  - Failure case analysis (wrong samples, weak classes, etc.)
  - WandB evidence (monitoring, config, overview, etc.)

Project 2: Leaderboard



- **Leaderboard will accept only one model per student.**
- 5/20 – 6/9: Professor and students may occasionally assign qualitatitve score. The model may change.
- 6/10 - 6/17: Qualitative scores will be reset, as the model should be fixed.

Project 2: Submission

Due: **6/9 23:59**

Submission

- Compress the entire codebase, **generator** checkpoint, and report into a single zip file.
- Do not need to submit the **discriminator**.
- Will announce how to submit your result.

- Important Notes
  - **Training data will be announced ASAP.**
  - **Submission site and base 256x256 will be announced in the 5/20 class.**

  - Report should include WandB log and Overview page capture.
  - We will check your WandB page during class, after project deadline.
  - If you are not sure of anything (code, task, performance, data, etc), ask professor anytime.

Project 2: Submission

(Zip file structure)
2025xxxxxx_project02.zip
- **src/**
- **checkpoints/**
  - model.pth
- 2025xxxxxx_project02_report.pdf
- pyproject.toml
- README.md

- README.md must include:
  - How to train and install dependencies.
  - How to generate the images from noise.
- Report
  - Report length must be no more than **6 pages**. Use an **11pt font size** for the main text. Submit the report in **PDF** format. There are no other formatting requirements.